

DATA SHEET

Three Phase Induction Motor - Squirrel Cage



Customer :

Product line : W22 NEMA Premium Efficiency Three-Phase Product code : 14124241

Frame	: 586/7T	Locked rotor time	: 28s (cold) 16s (hot)
Output	: 600 HP (440 kW)	Temperature rise	: 80 K
Poles	: 4	Duty cycle	: Cont.(S1)
Frequency	: 60 Hz	Ambient temperature	: -20°C to +40°C
Rated voltage	: 575 V	Altitude	: 1000 m.a.s.l.
Rated current	: 532 A	Protection degree	: IP55
L. R. Amperes	: 3777 A	Cooling method	: IC411 - TEFC
LRC	: 7.1x(Code G)	Mounting	: F-1
No load current	: 168 A	Rotation ¹	: Both (CW and CCW)
Rated speed	: 1790 rpm	Noise level ²	: 81.0 dB(A)
Slip	: 0.56 %	Starting method	: Direct On Line
Rated torque	: 1760 ft.lb	Approx. weight ³	: 4701 lb
Locked rotor torque	: 229 %		
Breakdown torque	: 250 %		
Insulation class	: F		
Service factor	: 1.00		
Moment of inertia (J)	: 236 sq.ft.lb		

Output	50%	75%	100%	Foundation loads	
Efficiency (%)	96.2	96.4	96.5	Max. traction	: 6831 lb
Power Factor	0.74	0.82	0.86	Max. compression	: 11532 lb

Losses at normative operating points (speed;torque), in percentage of rated output power

P1 (0,9;1,0)	P2 (0,5;1,0)	P3 (0,25;1,0)	P4 (0,9;0,5)	P5 (0,5;0,5)	P6 (0,5;0,25)	P7 (0,25;0,25)
4.3	3.3	2.6	2.7	1.8	1.5	0.8

	Drive end	Non drive end
Bearing type	: NU-322 C3	: 6319 C3
Sealing	: Taconite Labyrinth	: Taconite Labyrinth
Lubrication interval	: 4000 h	: 8000 h
Lubricant amount	: 60 g	: 45 g
Lubricant type	: Mobil Polyrex EM	

Notes

This revision replaces and cancel the previous one, which must be eliminated.

- (1) Looking the motor from the shaft end.
- (2) Measured at 1m and with tolerance of +3dB(A).
- (3) Approximate weight subject to changes after manufacturing process.
- (4) At 100% of full load.

These are average values based on tests with sinusoidal power supply, subject to the tolerances stipulated in NEMA MG-1.

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Customer :

Space heater information

Voltage: 200 - 240 V

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TORQUE AND CURRENT VS SPEED CURVE

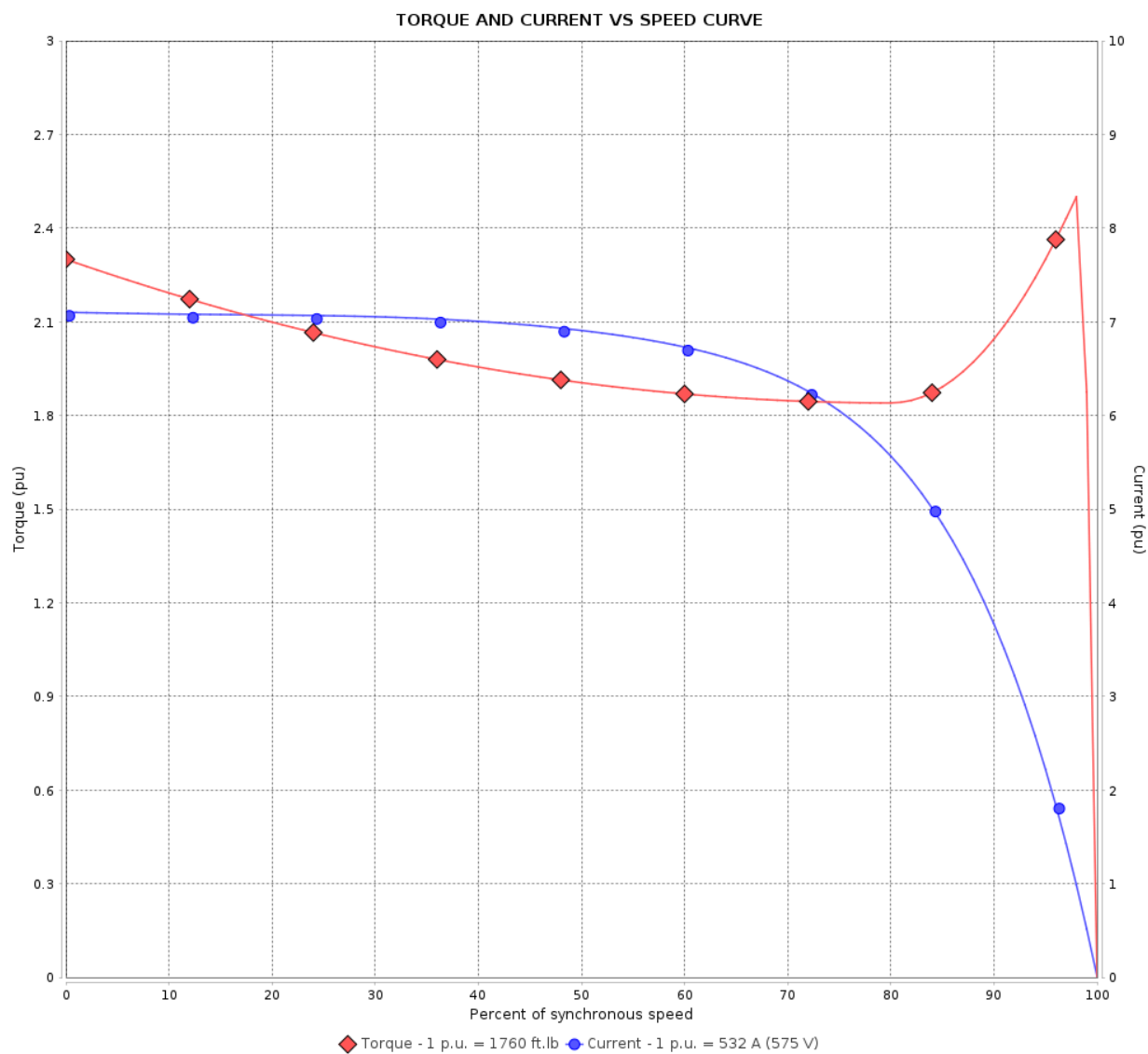
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Three-Phase

Product code : 14124241



Performance : 575 V 60 Hz 4P

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Rated torque	: 1760 ft.lb	Insulation class	: F
Locked rotor torque	: 229 %	Service factor	: 1.00
Breakdown torque	: 250 %	Temperature rise	: 80 K
Rated speed	: 1790 rpm		

Locked rotor time : 28s (cold) 16s (hot)

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LOAD PERFORMANCE CURVE

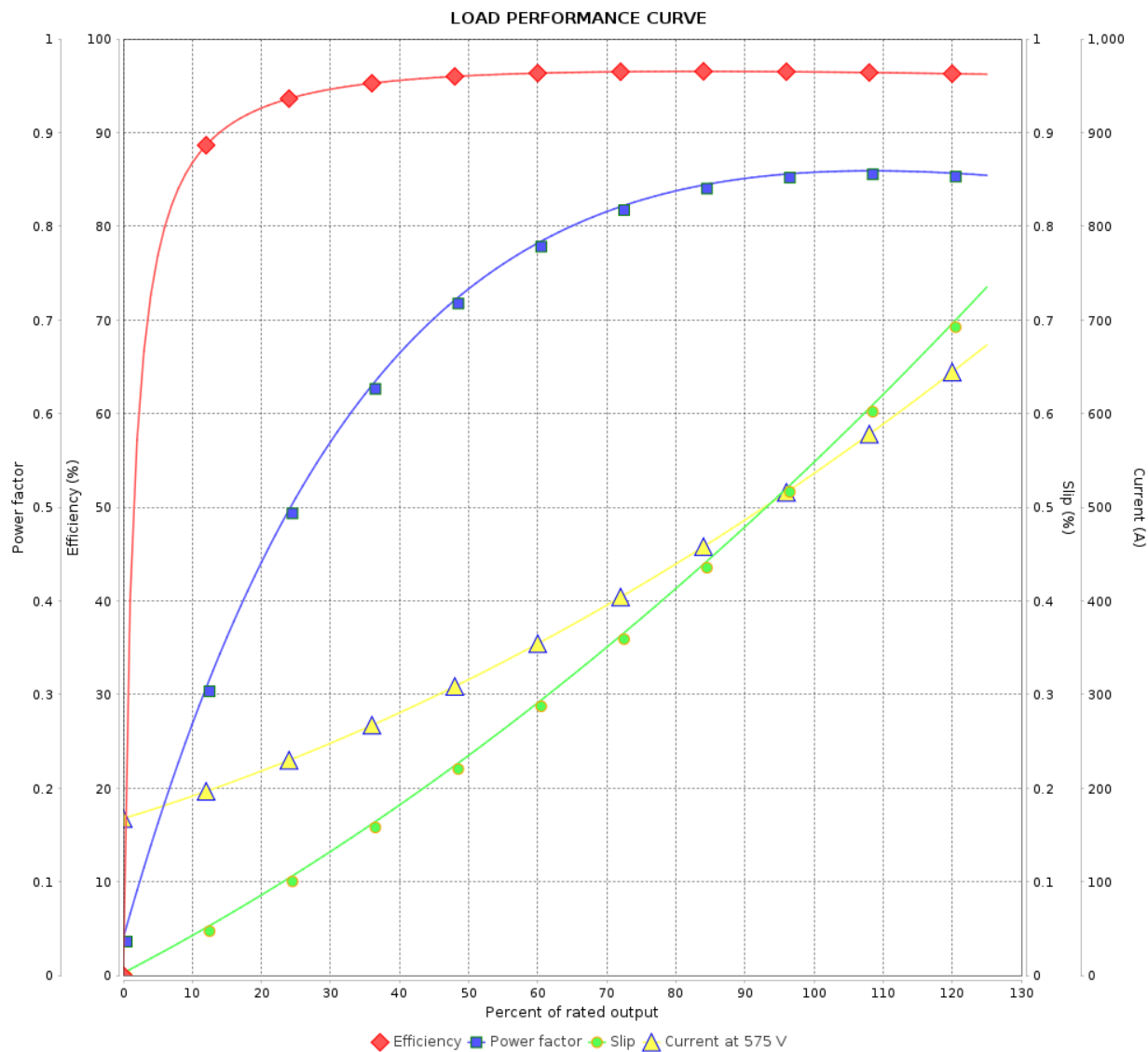
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THERMAL LIMIT CURVE

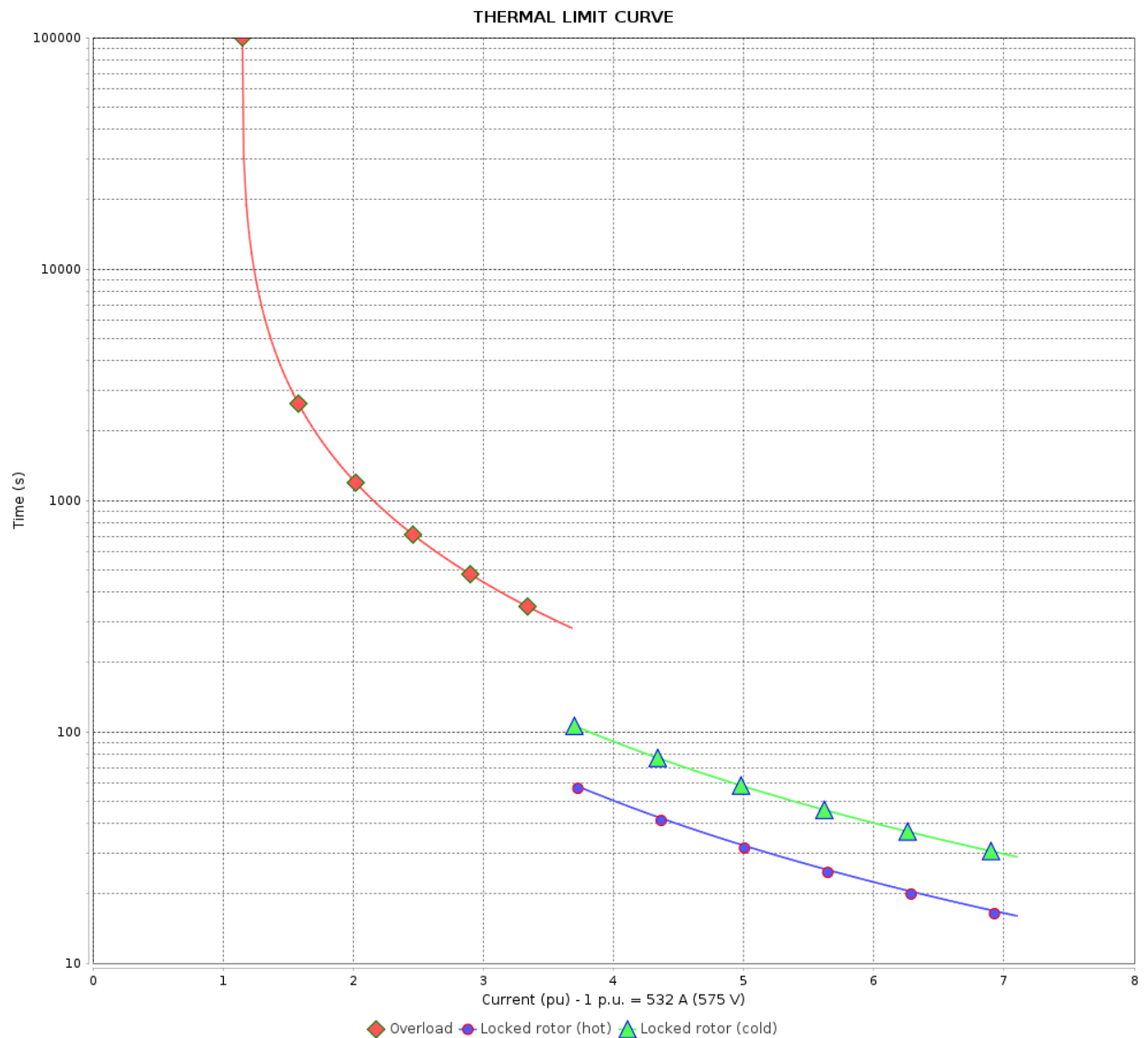
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Heating constant

Cooling constant

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VFD OPERATION CURVE

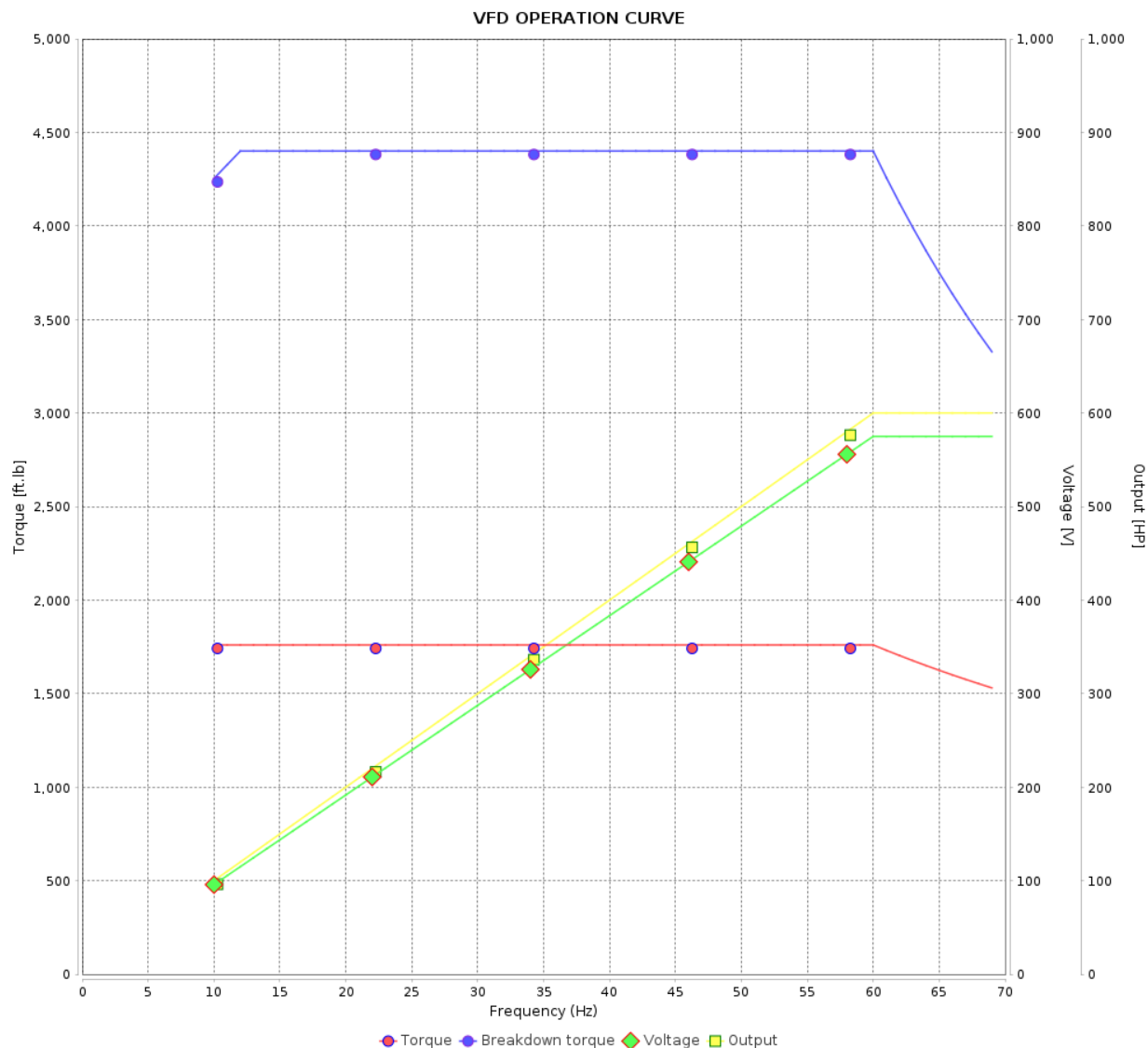
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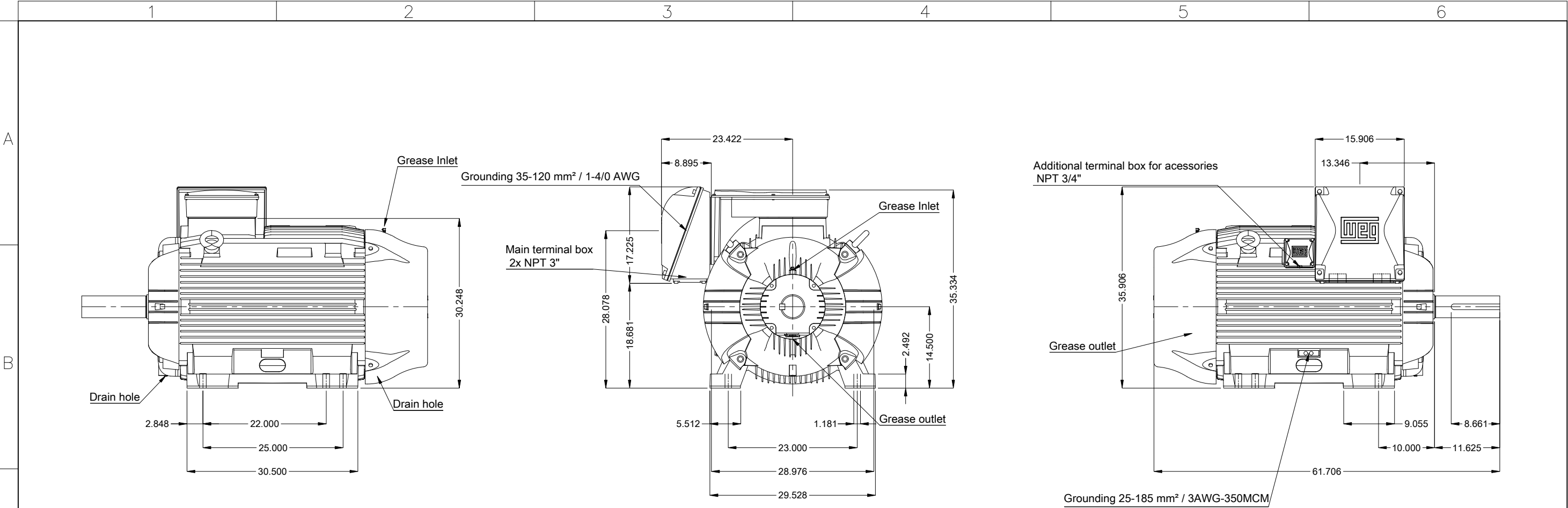


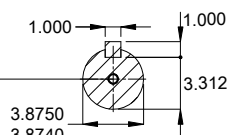
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D										
600 HP 04 Poles 60 Hz										
Without vertical jackscrews										
DE roller bearing										
Shaft locking device										
E	NDE endshield with electrically insulated bearing hub									
	Drive-End Air Deflector									
	Color RAL 5009									
	Painting plan 203A									
	Mounting NEMA F-1									
	Mounting B3R(D)									
	DE Shaft End  DUNC 7/8"-9 WEG WPR-7339									